Simple Network Management Protocol (SNMP): Architecture, Operation, and Practical Implementation Using GNS3

Abstract

Simple Network Management Protocol (SNMP) is a widely used application-layer protocol that enables centralized monitoring and management of network devices. It allows network administrators to monitor device status, collect performance information, and detect network issues efficiently without requiring direct access to each device. This report explores the fundamental concepts of SNMP, including its architecture, components, communication process, Management Information Base (MIB), Object Identifiers (OIDs), SNMP message types, and the different versions of the protocol.

To demonstrate the practical operation of SNMP, a network topology consisting of two routers was created using GNS3. One router was configured to function as the SNMP-enabled managed device (Agent), while the other was used to verify connectivity and demonstrate communication within the network. SNMP was configured on the managed router, and the implementation verified that device information could be monitored successfully in the simulated environment.

The study demonstrates that SNMP provides a simple and efficient solution for monitoring network devices and collecting operational data. It also highlights the advantages of SNMPv3 over earlier versions by offering authentication and encryption features, making it a more secure choice for modern enterprise networks.

Introduction

Modern computer networks form the backbone of communication in businesses, educational institutions, government organizations, and cloud-based services. As networks continue to expand in size and complexity, managing and monitoring network devices efficiently has become a critical task for network administrators. Manual monitoring of each router, switch, or server is time-consuming and impractical, especially in large-scale environments. This challenge has led to the development of network management protocols that enable centralized monitoring and control of network devices.

Simple Network Management Protocol (SNMP) is one of the most widely adopted protocols for network management. Operating at the Application Layer of the TCP/IP model, SNMP provides a standardized method for collecting information from network devices, monitoring their performance, detecting faults, and, when authorized, modifying configuration parameters remotely. It follows a Manager-Agent architecture, where an SNMP Manager communicates with SNMP Agents running on managed devices to retrieve operational data such as CPU utilization, memory usage, interface statistics, and device uptime.

The motivation for choosing this topic is its practical importance in modern networking. SNMP is widely implemented in enterprise networks, Internet Service Providers (ISPs), and data centers to ensure continuous monitoring and efficient management of network infrastructure. Understanding SNMP provides valuable knowledge of how network administrators maintain the health and availability of communication systems while reducing downtime and improving operational efficiency.

As part of this study, a practical implementation was carried out using GNS3, where a simple network topology consisting of two routers was configured to demonstrate the working principles of SNMP. This hands-on implementation reinforces the theoretical concepts and provides practical insight into how SNMP operates in a simulated network environment.

The study of SNMP is highly relevant in the field of Computer Networks because it introduces the concepts of centralized network management, automated monitoring, fault detection, and performance analysis. These concepts are essential for designing, operating, and maintaining reliable and scalable network infrastructures in today's interconnected world.

Objective

The primary objective of this report is to study and understand the working principles of the **Simple Network Management Protocol (SNMP)** and its role in modern computer networks. The report aims to explain the architecture, components, communication process, and different versions of SNMP, highlighting how the protocol enables centralized monitoring and management of network devices.

The specific objectives of this work are:

- To understand the purpose and importance of SNMP in network management.
- To study the architecture of SNMP, including the roles of the SNMP Manager, SNMP Agent, Managed Devices, Management Information Base (MIB), and Object Identifiers (OIDs).
- To explain the various SNMP message types, such as GET, GETNEXT, GETBULK, SET, RESPONSE, and TRAP.
- To compare the features and security aspects of SNMPv1, SNMPv2c, and SNMPv3.
- To demonstrate the practical implementation of SNMP using **GNS3** by configuring a simple network topology consisting of **two routers**.
- To verify SNMP communication and observe how device information can be monitored within the simulated network environment.
- To understand the practical applications and advantages of SNMP in enterprise network monitoring and management.

By achieving these objectives, this report provides both theoretical knowledge and practical experience, demonstrating how SNMP simplifies network administration and contributes to efficient monitoring, troubleshooting, and maintenance of computer networks.

Why do we need SNMP?

To understand the importance of SNMP, it is helpful to look back at the 1970s and 1980s, when computer networks were managed manually using the Command Line Interface (CLI) on individual devices. At that time, there were no centralized Network Management Systems (NMS) or automated methods for collecting network data. This made monitoring, troubleshooting, and analyzing network issues a slow and difficult process. In most cases, network administrators became aware of problems only after users reported service disruptions, leading to increased downtime and inefficient network management.

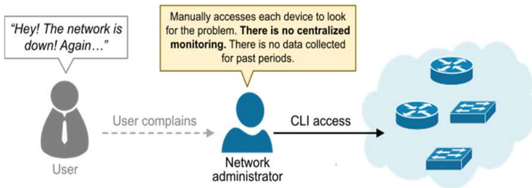

Figure 1 Networks in the 1980s - networkacademy.io

As computer networks grew larger and became essential for businesses in the late 1980s, manual network management was no longer practical. Without a centralized monitoring system, administrators could only detect problems after users reported them or during routine device checks. This highlighted the need for a standardized network management protocol that could continuously monitor devices, collect network statistics, and notify administrators of faults in real time. This demand ultimately led to the development of SNMP.

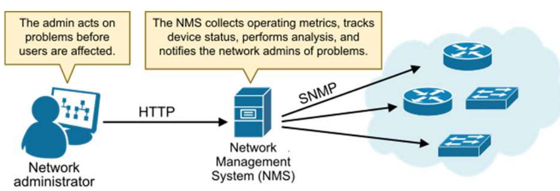

Figure 2A network with a NMS - networkacademy.io

The idea behind SNMP

The idea behind SNMP is that all devices connected to the network, such as a computer, router, switch, firewall, printer, access point, etc., **share the same characteristics**. They all have CPU, RAM, DISC, OS, interfaces, buffers, hostname, IP addresses, temperature sensors, etc. Hence, if every device characteristic can be represented as a variable, an application can monitor and manage the device by reading and modifying these variables. So that's what the protocol does; **it represents each device's characteristic as an object** identified by an OID (Object Identifier), as shown in the diagram below.

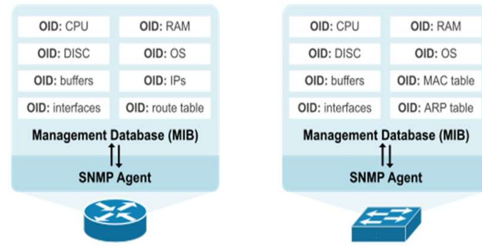

Figure 3 SNMP variables - networkacademy.io

It then organizes the OIDs hierarchically into a structured database called the Management Information Base (MIB). Later, we will discuss OIDs and the MIB in more detail.

The SNMP components

An SNMP-managed network includes three main components, as shown in the diagram below: an SNMP manager, SNMP agents, and a Management Information Base (MIB).

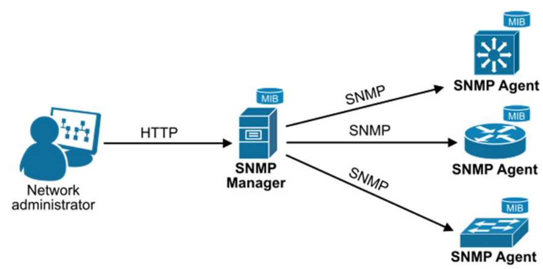

Figure 4 SNMP Components - networkacademy.io

The SNMP manager acts as the interface between the network administrator and the management system. It is also referred to as a Network Management Station (NMS). It collects operational data from devices using SNMP query messages.

The SNMP agent is a software module installed in the managed device's operational system (OS). It maintains information in the device's **MIB database** and allows the SNMP manager to read and write the values stored there.

The Management Information Base (MIB) is the database of the device's variables (OIDs) that can be managed. Typically, an MIB is unique to a device.

The MIB Hierarchy

Each SNMP agent (technically each managed device) has its own Management Information Base (MIB), which defines the variables that the agent can read and write. Each available variable on the device's MIB is called an Object and is identified by an Object Identifier (OID).

The MIB structure is organized as a tree hierarchy, where each node is identified by a name or number, as shown in the diagram below.

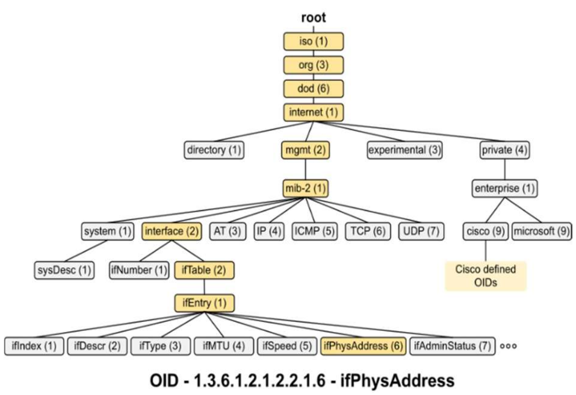

Figure 5 SNMP MIB - networkacademy.io

The first few branches of the MIB tree, such as iso.org.dod.internet.mgmt, are always the same because they follow a standardized hierarchy defined by ISO. This structure ensures a consistent, globally recognized framework for organizing objects across different systems and vendors. Starting with iso (1), the tree branches into org (3), representing organizations, and then into dod (6) for the Department of Defense, which initially managed the internet standards (back in the 1980s-1990s). From there, internet (1) branches into areas like mgmt (2) for managed objects and private (4) for vendor-specific objects. This standardized organization allows for interoperability and uniformity across all MIB implementations.

Working directly with MIBs can be challenging due to the complexity of the variable names and numbers. The NMS tools, such as Cisco Prime, PRTG, Solarwinds, etc., simplify this process **by hiding the detailed MIB structures under the hood**. Network administrators typically don't need to know the MIB structure of the devices they manage. However, a general understanding of the MIB structure and how it works is good to have, especially if you are taking a CCNA/CCNP/CCIE exam.

Notice that the name notation is designed for humans. **Machines use number notation.**

SNMP messages

SNMP is a request-response protocol. The network management system (NMS) sends out a request, and the remote device returns a response. However, since the status of an object can change over time (for example, an interface can be up now but down in 5 minutes), the NMS queries the device every couple of minutes to retrieve updated information. This is referred to as the polling interval, as shown in the diagram below.

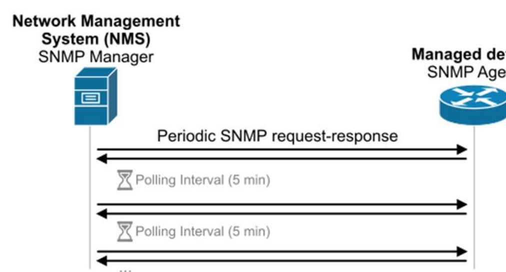

Figure 6 SNMP polling - networkacademy.io

This request-response communication uses three message categories: Get, Set, and Trap. The messages contain a header and a PDU (protocol data unit). The headers consist of the community string used as an authentication (more later on). The PDU depends on the type of message being sent but typically consists of the requested information.

Get messages: The SNMP manager (called the NMS) uses SNMP Get messages to retrieve the value of a specific managed object identified by its OID (Object Identifier) from an SNMP agent.

- **SNMP Get:** Retrieves a single, specific OID.

- **SNMP GetNext:** Retrieves the next OID in the MIB tree.
- **SNMP GetBulk:** Retrieves multiple objects in a single request for efficient bulk data collection.

Set messages: SNMP set messages are used by a network management system (NMS) to modify or configure values on a managed device, such as changing an interface's status or updating device settings. These messages contain the desired Object Identifier (OID) and the new value to apply to the device. The device processes the request and, if successful, updates the configuration or state accordingly, sending a response back to the NMS for confirmation.

Notice that SNMP set messages require authentication and RW access to ensure that only authorized users can modify a device's configuration. The authentication mechanism depends on the SNMP version in use:

- **SNMPv1 and SNMPv2**: Use a community string as a simple password for authentication. If the community string matches the device's configured string, the request is accepted.
- **SNMPv3**: Provides more robust security, including authentication with usernames and optional encryption, ensuring confidentiality and integrity of the messages.

Trap messages: One of SNMP's limitations is its request-response nature. This implies that a device only reports its operation status when a query from the NMS is made. This is not efficient when there is an event on the device that needs the attention of the network administrators, such as an interface going down. That's why the protocol implements real-time notifications called SNMP Traps.

An SNMP trap is an unsolicited message sent by an SNMP agent (a device) to an SNMP manager (the NMS) to notify it of specific events or conditions on the managed device. Unlike the typical request-response flow, traps are one-way communications initiated by the agent.

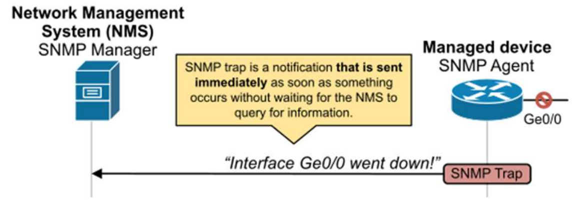

Figure 7 SNMP Trap - networkacademy.io

SNMP Traps help identify issues like hardware failures or threshold breaches in real-time. For example, an interface goes down. The network devices immediately send an SNMP trap to the NMS. A network administrator immediately takes action to prevent problem escalation or outage

Configuring SNMP on Cisco devices

Now let's be a bit more practical and show how we configure basic SNMP functionalities on Cisco devices.

Enabling SNMP on a Cisco router

Enabling SNMP on a Cisco router is as simple as adding one or two commands under global configuration mode as shown in the output below.

```
Router(config)# snmp-server community [community-string] [permissions]
```

We only need to specify the community and the permission level.

What is the SNMP community?

An SNMP community is **a password-like string used to control access to an SNMP-enabled device**. It acts as a basic form of authentication for SNMP operations and determines what level of access a user or network management system (NMS) has to the device. There are typically two types of community strings:

- **Read-Only (RO):** This community allows the SNMP manager to retrieve only data from the device, such as checking performance metrics or configurations, without being able to modify any settings.
- **Read-Write (RW):** This community allows the SNMP manager to retrieve data from the device and modify its configuration or settings.

Community strings act as simple authentication keys for SNMP operations. They are sent in clear text over the network, which is a security concern in environments requiring higher security. In more secure setups, SNMPv3 is often preferred since it supports encryption and stronger authentication mechanisms.

The following examples show how to configure read-only and read-write communities.

```
Router(config)# snmp-server community MyPublicCommunity ro

Router(config)# snmp-server community MyPrivateCommunity rw
```

These two commands are enough to enable the SNMP functionality on the device and add it to your NMS system that monitors and manages the network.

Enable SNMP Traps (Optional)

In real-world deployments, we also want to enable the SNMP trap functionality that notifies the NMS immediately when an event occurs on the device. We enable it using the following command.

```
Router(config)# snmp-server host [nms-ip] version 2c [community-string]

Router(config)# snmp-server enable traps
```

Notice that the NMS-ip is the IP address of the management system, and the community string is the password.

Implementation

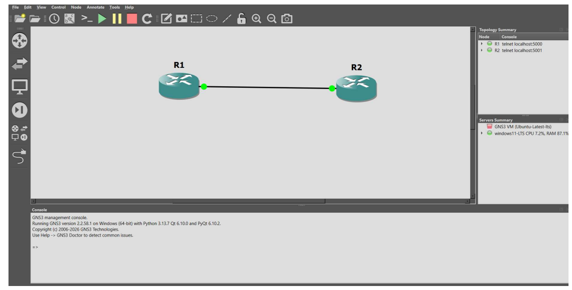

Figure 8 Router setup

R1 - SNMP MANAGER

R2 - SNMP AGENT

```
R1#configure ter
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#snmp-server community public rw
R1(config)#exit
R1#
*Mar  1 00:04:15.871: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

Figure 9 R1 - SNMP manager

```
R2#conf
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#snmp-server community public ro
R2(config)#exit
R2#
*Mar  1 00:03:40.731: %SYS-5-CONFIG_I: Configured from console by console
R2#
```

Figure 10 R2 - SNMP agent

Set ip iddress for routers

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#inf
R1(config)#interface fas
R1(config)#interface fastEthernet 0/1
R1(config-if)#ip address 10.0.0.1 255.0.0.0
R1(config-if)#no shutdown
R1(config-if)#
*Mar  1 00:09:15.927: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar  1 00:09:16.927: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R1(config-if)#exit
R1(config)#exit
R1#sho
*Mar  1 00:09:22.343: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES unset  administratively down down
FastEthernet0/1        10.0.0.1        YES manual up                    up
R1#
```

Figure 11 Assigning IP for interface of R1

```
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#interface fa
R2(config)#interface fastEthernet 0/0
R2(config-if)#ip address 10.0.0.2 255.0.0.0
R2(config-if)#no shutdown
R2(config-if)#e
*Mar  1 00:06:59.331: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar  1 00:07:00.331: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to
R2(config-if)#exit
R2(config)#exit
R2#show ip
*Mar  1 00:07:05.539: %SYS-5-CONFIG_I: Configured from console by console
R2#show ip interface brief
Interface                  IP-Address      OK? Method Status                Protocol
FastEthernet0/0            10.0.0.2        YES manual up                    up
FastEthernet0/1            unassigned      YES unset  administratively down down
R2#
```

Figure 12 Assigning IP for interface of R2

Check how many interfaces R2 has:

```
R1(config)#snmp-server manager
R1(config)#exit
R1#
*Mar  1 00:17:20.019: %SYS-5-CONFIG_I: Configured from console by conso
R1#snmp get v2c 10.0.0.2 public oid 1.3.6.1.2.1.2.1.0
SNMP Response: reqid 1, errstat 0, erridx 0
 ifNumber.0 = 3
R1#snmp get v2c 10.0.0.2 public oid 1.3.6.1.2.1.2.2.1.2.1
SNMP Response: reqid 2, errstat 0, erridx 0
 ifDescr.1 = FastEthernet0/0
R1#
```

Conclusion and Future Work

Conclusion

This report presented a comprehensive study of the Simple Network Management Protocol (SNMP), covering its history, architecture, components, communication process, and different protocol versions. The study demonstrated how SNMP enables centralized monitoring and management of network devices, making it an essential protocol for maintaining reliable and efficient computer networks.

A practical implementation was carried out using GNS3, where a simple topology consisting of two routers was configured to demonstrate SNMP communication. The implementation verified that SNMP can successfully monitor network devices and retrieve management information in a simulated environment. This practical exercise reinforced the theoretical concepts and highlighted the importance of SNMP in simplifying network administration, fault detection, and performance monitoring.

Overall, the study shows that although SNMP has evolved over the years, it remains one of the most widely used protocols for network monitoring. With the enhanced security features offered by SNMPv3, it continues to play a vital role in modern enterprise and service provider networks.

Future Work

The current implementation focused on demonstrating the basic functionality of SNMP using two routers in a GNS3 environment. Future work can extend this project by:

- Implementing SNMP in a larger network topology with multiple routers and switches.
- Integrating an SNMP Manager such as LibreNMS, PRTG, or Zabbix for real-time monitoring and graphical visualization.
- Configuring and testing SNMPv3 to evaluate its authentication and encryption features.
- Demonstrating SNMP Trap notifications for automatic fault detection and alert generation.
- Monitoring additional network parameters such as CPU utilization, memory usage, interface statistics, and bandwidth consumption.
- Comparing SNMP with modern network management approaches to understand their advantages and limitations.

These enhancements would provide a more comprehensive understanding of SNMP and its practical applications in enterprise network management.

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